

Brand Owner / Head Office → Reports & Analytics Module

System Walkthrough, Architectural Specification & Implementation Traceability Matrix

Executive Architectural Summary: The Reports & Analytics module acts as the centralized business intelligence layer for the Brand Owner, integrating cross-module telemetry across Operations, Revenue, Staff Productivity, Client Retention, Inventory Consumption, Marketing Campaign ROI, and Franchise Performance Scorecards. It deliberately avoids duplicating operational workflows, focusing strictly on high-level multi-branch benchmarking, trend analytics, custom report builders, and multi-format exports (CSV, Excel, PDF).

1. Official PRD Traceability Matrix

Mapping all official product requirements to their concrete implementation within the codebase:

PRD REQUIREMENT ID	FEATURE NAME	PRODUCT REQUIREMENT DESCRIPTION	IMPLEMENTATION DETAILS IN CODE
SALO-PRD-083	Executive Overview Dashboard	High-level business velocity, 8 comparative KPI cards, and cross-module intelligence.	ExecutiveOverviewTab.tsx: 8 KPI cards with previous period deltas, 7 visual trend cards, and multi-branch comparison table.
SALO-PRD-084	Operations & Occupancy	Bookings, completions, cancellations, no-shows, walk-ins, waitlist, and workstation occupancy.	OperationsAnalyticsTab.tsx: 7 operational KPIs, slot occupancy pacing bars, booking source channels, and branch comparison matrix.
SALO-PRD-085	Revenue & Collections	Services, retail sales, packages, memberships, tax, discounts, refunds, and net turnover.	RevenueAnalyticsTab.tsx: 9 PRD revenue measures, ritual category splits, payment tender methods, and branch revenue ledger.
SALO-PRD-086	Staff Productivity & Scorecard	Utilisation, completed services, retail upsell, rebooking rates, ratings, targets, and commission.	StaffPerformanceAnalyticsTab.tsx: Stylist performance matrix, target achievement badges, and commission accruals.
SALO-PRD-087	Client Retention & Cohorts	New vs repeat ratio, rebooking conversion, inactive cohorts, frequency, LTV, and churn risk.	ClientRetentionAnalyticsTab.tsx: 5 cohort segments (Champions, Regulars, New, Inactive, At Risk) with frequency and spend metrics.
SALO-PRD-088	Inventory & Recipe Yield	Stockholding value, consumption, wastage discrepancies, expiring stock, and recipe vs actual.	InventoryAnalyticsTab.tsx: 6 inventory KPIs, recipe vs actual yield variance ledger, and branch stock valuation breakdown.
SALO-PRD-089	Campaign ROI & Conversion	Audience reach, delivered rate, bookings, attendance, attributed revenue, coupon usage, and opt-outs.	MarketingAnalyticsTab.tsx: Multi-channel campaign ROI ledger, conversion multipliers, and opt-out tracking.
SALO-PRD-090	Franchise Scorecard	Territory partner revenue, chair utilisation, customer CSAT ratings, royalty fees, and compliance.	FranchiseAnalyticsTab.tsx: Partner scorecard table, regional distribution benchmarking, and 10% royalty pool tracking.

PRD REQUIREMENT ID	FEATURE NAME	PRODUCT REQUIREMENT DESCRIPTION	IMPLEMENTATION DETAILS IN CODE
SALO-PRD-091	Custom Report Builder	Dynamic report builder with domain module, primary dimensions, and calculated measures.	CustomReportsTab.tsx: Interactive builder panel with live table preview canvas and instant library saving.
SALO-PRD-092	Saved Reports Library	Repository of custom saved queries, automated schedules, duplication, and export actions.	CustomReportsTab.tsx: Saved reports table with status badges (Scheduled, Generated, Saved) and query duplication.
SALO-PRD-093	Universal Multi-Format Export	Portaled export engine supporting CSV, Excel (.xlsx), and print-ready PDF dossiers.	UniversalExportModal.tsx: Full-screen portaled modal (z- [9999]) with custom column picker and date range filters.

2. Business Rationale & Why We Implemented Everything

A detailed rationale explaining the commercial need and architectural choices for each capability:

1. Why Centralized Analytics without Operational Workflows? Operational execution (editing shifts, managing inventory, booking appointments) belongs in specialized modules. Centralized Analytics gives the Brand Owner a holistic 360° overview across all 6 branches without operational clutter.	2. Why Specific PRD Revenue Stream Measures? Salon businesses operate on blended revenue streams. Separating in-salon services (68.8%) from retail products (12.9%), package advances (10.4%), and recurring memberships (7.4%) gives leadership actionable cashflow visibility.
3. Why Recipe vs Actual Yield Variance Ledger? Chemical over-dispensing (hair color, developer, facial serums) causes direct margin leakage. Tracking standard recipe doses against actual bowl dispensing enforces stylist discipline and reduces wastage to <0.4%.	4. Why Cohort Segmentation with Churn Risk? Customer retention is 5x more cost-effective than new acquisition. Identifying 1,240 dormant clients (60-90 days lapsed) triggers targeted win-back campaigns before they become permanent churn risks.
5. Why Multi-Format Universal Export Engine? Different executive stakeholders require specific formats: finance teams require raw CSV/Excel for reconciliation, while board members and auditors need styled executive PDF dossiers.	6. Why Lightweight Custom Report Builder? Eliminates the complexity of heavy external BI tools while giving Brand Owners instant flexibility to mix modules (Revenue, Staff, Inventory) with primary grouping dimensions (Branch, Date, Category).

3. UI Architecture & Page Navigation Flow

The Reports & Analytics module is structured under a unified responsive master container with persistent sub-navigation:

- **Executive Overview (/reports?tab=overview):** 8 comparative KPI cards, monthly revenue velocity bars, client ratio widgets, and branch ranking.
- **Operations (/reports?tab=operations):** 7 appointment KPIs, weekly slot occupancy pacing, and origin channel distribution (App 48%, Call Center 22%).
- **Revenue (/reports?tab=revenue):** 9 PRD revenue measures, service category split, payment tender methods, and multi-branch revenue matrix.
- **Staff Performance (/reports?tab=staff):** Stylist productivity scorecard, workstation utilisation, rebooking rates, and commission accruals.
- **Client & Retention (/reports?tab=clients):** 5 client cohort segments, visit frequency rhythm, and churn risk alerts.
- **Inventory (/reports?tab=inventory):** Stock valuation, consumable usage, wastage discard, and recipe vs actual yield matrix.
- **Marketing (/reports?tab=marketing):** Campaign reach, direct bookings captured, attendance rates, and ROI multipliers.
- **Franchise (/reports?tab=franchise):** Licensed partner performance scorecard, chair utilisation, and 10% royalty pool tracking.
- **Custom Reports (/reports?tab=custom):** Sub-tab toggle between the Custom Report Builder and the Saved Reports Library.